Greenwash

Testing Artifact

(SENG 401 L02 Group 02)

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Testing Overview

The purpose of testing is to verify that the Greenwash game functions correctly, fairly, and consistently across its core systems:

- Gameplay logic (roles, turns, voting, win conditions)
- State transitions (phases of the game)
- Database interactions (player/session data)
- Input handling (user actions like voting, joining)

Testing was conducted using a combination of:

- Unit Testing (individual components)
- System Testing (end-to-end gameplay)
- Exploratory Testing (real player interaction scenarios)

Test Cases

Lobby and Setup:

TC001	**Player Creating Lobby**	
Preconditions	**The player must be on the menu page**	
Procedure	**Action**	**Expected Result**
1	Enter player name	The player name is shown in the respective field
2	Click on the “create session” button	The screen says “Creating a new session for (player name)...”
Expected Result	The user is taken to the game and a unique 6-character lobby code is generated	
Actual Result	Lobby created with valid code	
Postcondition	**The player is on the game page and is given a character selection screen**	
Status	Pass	

TC002	Player Creating Lobby With Empty Player Name	
Preconditions	**The player must be on the menu page**	
Procedure	**Action**	**Expected Result**
1	Click on the “create session” button	The player is told “Enter your player name first”
Expected Result	The player should be told “Enter your player name first”	
Actual Result	The player is told “Enter your player name first”	
Postcondition	**The player is still on the menu page**	
Status	Pass	

TC003	Player Joining Lobby	
Preconditions	**The player must be on the menu page and a lobby with the corresponding code must exist.**	
Procedure	**Action**	**Expected Result**
1	Enter valid lobby code	The lobby code is shown in the respective field
2	Enter player name	The player name is shown in the respective field
3	Click on the “join session” button	The screen says “Joining (lobby code) as (player name)”
Expected Result	The player should be given entry to the lobby from the corresponding code	
Actual Result	The player is given entry to the lobby from the corresponding code	
Postcondition	**The player is on the game page and is given a character**	

	selection screen. The player is added to the current player list on the screen.
Status	Pass

TC004	**Player Joining Lobby With Empty Lobby Code**	
Preconditions	**The player must be on the menu page and a lobby with the corresponding code must exist**	
Procedure	**Action**	**Expected Result**
1	Enter player name	The player name is shown in the respective field
2	Click on the “join session” button	The player is told “Enter both a lobby code and player name”
Expected Result	The player should be forced to enter a lobby code	
Actual Result	The player is forced to enter a lobby code	
Postcondition	**The player is still on the menu page**	
Status	Pass	

TC005	**Player Joining Lobby With Empty Player Name**	
Preconditions	**The player must be on the menu page and a lobby with the corresponding code must exist**	
Procedure	**Action**	**Expected Result**
1	Enter valid lobby code	The lobby code is shown in the respective field
2	Click on the “join	The player is told “Enter both a lobby

	session” button	code and player name”
Expected Result	The player should be forced to enter a player name	
Actual Result	The player is forced to enter a player name	
Postcondition	**The player is still on the menu page**	
Status	Pass	

TC006	**Player Joining Lobby With Invalid Lobby Code**	
Preconditions	**The player must be on the menu page and a lobby with the corresponding code must exist.**	
Procedure	**Action**	**Expected Result**
1	Enter invalid lobby code (code that isn’t actively being used by a lobby)	The lobby code is shown in the respective field
2	Enter player name	The player name is shown in the respective field
3	Click on the “join session” button	The player is told “Lobby not found”
Expected Result	The player should be told the lobby does not exist, indicating that a valid code should be entered	
Actual Result	The player is told the lobby does not exist, indicating that a valid code should be entered	
Postcondition	**The player is still on the menu page.**	
Status	Pass	

TC007	Player Joining Lobby With Taken Player Name	
Preconditions	**The player must be on the menu page and a lobby with the corresponding code must exist.**	
Procedure	**Action**	**Expected Result**
1	Enter valid lobby code	The lobby code is shown in the respective field
2	Enter player name being used by another player currently in the lobby	The player name is shown in the respective field
3	Click on the “join session” button	The player is told “Username already taken in lobby”
Expected Result	The player should be told the player name is already being used, indicating that another name should be entered	
Actual Result	The player is told the player name is already being used, indicating that another name should be entered	
Postcondition	**The player is still on the menu page**	
Status	Pass	

TC008	Character Selection	
Preconditions	**The player must have joined or created a lobby**	
Procedure	**Action**	**Expected Result**
1	Select one of the available avatars	The avatar is indicated as selected
2	Click on the “confirm” button.	The character selection screen is closed
Expected Result	The player’s avatar should be broadcasted to all of the other players in the lobby	

Actual Result	The player's avatar is broadcasted to all of the other players in the lobby
Postcondition	**The player is still on the game page**
Status	Pass

TC009	**Minimum Player Requirement**	
Preconditions	**The player must have joined or created a lobby and have less than 5 people in the lobby**	
Procedure	**Action**	**Expected Result**
1	Select one of the available avatars	The avatar is indicated as selected
2	Click on the "confirm" button	The character selection screen is closed
Expected Result	The game should not start	
Actual Result	The game did not start	
Postcondition	**The player is still on the game page**	
Status	Pass	

TC010	**Auto Game Start**	
Preconditions	**The player must have joined or created a lobby and have more than 5 people in the lobby (the other 4 already having their character selected)**	
Procedure	**Action**	**Expected Result**

1	Select one of the available avatars	The avatar is indicated as selected
2	Click on the “confirm” button	The character selection screen is closed
Expected Result	The game should automatically start and the player has been assigned their role	
Actual Result	The game automatically starts and the player has been assigned their role	
Postcondition	**The player is in the game**	
Status	Pass	

Roles & Initialization:

TC011	**Role Assignment Distribution**
Preconditions	**The game must have started**
Expected Result	The game should have assigned two people as exploiters and three people as reformers
Actual Result	The game assigned two people as exploiters and three people as reformers
Postcondition	**The player is shown their role, and has an option to press “okay”**
Status	Pass

TC012	Exploiter Identification	
Preconditions	**The roles must have been assigned to players**	
Procedure	**Action**	**Expected Result**
1	Click on the “okay” button	The role assignment screen is closed
Expected Result	Reformers should not be able to other’s roles, while exploiters should be able to identify the other exploiter	
Actual Result	Reformers can not see other’s roles, and exploiters can see who the other exploiter is	
Postcondition	**The player waits for the other to confirm**	
Status	Pass	

TC013	Role Acknowledgment	
Preconditions	**The roles must have been assigned and the 4 other players have not confirmed their roles**	
Procedure	**Action**	**Expected Result**
1	Click on the “okay” button	The role assignment screen is closed
Expected Result	The game should pause until all 5 players have confirmed their roles (pressed “okay”)	
Actual Result	The game pauses until all 5 players have confirmed their roles (pressed “okay”)	
Postcondition	**The gameplay does not start**	
Status	Pass	

Turn & Election System:

TC014	Leader Rotation	
Preconditions	**Every player has confirmed their rules and the nomination for vice has begun**	
Procedure	**Action**	**Expected Result**
1	The initial leader nominates a vice	The players are given a voting option (“Aye” or “Nay”)
2	All players vote for “Nay”	The nominee is rejected as vice
Expected Result	The game should move onto the next leader, in a fixed order, and begin the nomination process again	
Actual Result	The game moves onto the next leader, in a fixed order, and begins the nomination process again	
Postcondition	**The next leader nominates a vice**	
Status	Pass	

TC015	Vice Nomination Constraints	
Preconditions	**The vice nomination process has been done once.**	
Procedure	**Action**	**Expected Result**
1	The current leader must nominate their vice	The potential candidates are displayed
Expected Result	The game should not allow the current leader to pick either themselves, the previous vice, or the previous leader	
Actual Result	The options for the previous leader and previous vice are blocked out (saying “restricted”), and the current leader is not visible	
Postcondition	**The current leader nominates a vice**	

Status	Pass

TC016	**Election Voting**	
Preconditions	**The current leader is in the vice nomination process**	
Procedure	**Action**	**Expected Result**
1	The current leader nominates a player for vice	Every player must choose either “Aye” or “Nay” on whether the nominee should be elected
2	The player chooses “Aye”	The player’s voting screen closes.
Expected Result	The game should allow the player to cast only one secret vote and close the voting screen	
Actual Result	The other players are not informed about the player’s vote, and the player’s voting screen is closed	
Postcondition	**The player waits for everyone to cast their vote**	
Status	Pass	

TC017	**Election Result Broadcasting**	
Preconditions	**The 4 other players have cast their vote (“Aye”)**	
Procedure	**Action**	**Expected Result**
1	The player chooses “Aye”	The player’s voting screen closes
Expected Result	The game should show the tally of votes and whether or not the nominee has been elected vice	
Actual Result	The game shows the tally of votes (5-0) and that the nominee has been elected (“vote was successful”)	
Postcondition	**The elected vice is given policies to review**	
Status	Pass	

TC018	Election Tracker	
Preconditions	**2 consecutive vice election were unsuccessful**	
Procedure	**Action**	**Expected Result**
1	The current leader nominates a vice	Every player must choose either “Aye” or “Nay” on whether the nominee should be elected
2	Every player chooses “Nay”	The election results are shown
Expected Result	The game should automatically enact the policy card at the top of the deck	
Actual Result	The game automatically enacts the policy card at the top of the deck	
Postcondition	**The game is paused until everyone presses “next round”**	
Status	Pass	

Policy Mechanics:

TC019	Leader Discard Phase	
Preconditions	**The current leader has nominated a vice**	
Procedure	**Action**	**Expected Result**
1	Every player chooses “Aye”	The election is shown to be successful
Expected Result	The leader should be given a choice of 3 policy cards, and must discard 1 of them	
Actual Result	The leader is given a choice of 3 policy cards, and must discard 1 of them	
Postcondition	**The vice waits until the leader has decided**	

Status	Pass

TC020	**Vice Enact Phase**	
Preconditions	**The current leader is choosing which policy to discard**	
Procedure	**Action**	**Expected Result**
1	The leader chooses which policy to discard	The vice is shown the remaining 2 policies, and must choose which one to enact
2	The vice chooses one of the policies (sustainable)	The selection screen closes
Expected Result	The players should be shown if the policy enacted was sustainable, and the sustainable counter should increase	
Actual Result	The players are shown that the policy enacted was sustainable, and the sustainable counter increases	
Postcondition	**The game is paused until everyone presses “next round”**	
Status	Pass	

TC021	**Deck Management**	
Preconditions	**There is an insufficient amount of cards from the draw pile**	
Procedure	**Action**	**Expected Result**
Expected Result	The cards from the discard pile should be reshuffled into the draw pile	
Actual Result	The cards from the discard pile are reshuffled into the draw pile and the deck is restored	
Postcondition	**The game continues**	
Status	Pass	

TC022	Card Data Loading	
Preconditions	**The game has started**	
Procedure	**Action**	**Expected Result**
Expected Result	The cards should be loaded from the database properly and placed in the pile	
Actual Result	The cards are loaded from the database properly and placed in the pile	
Postcondition	**The game continues**	
Status	Pass	

Game Progression:

TC023	Win Condition – Reformers	
Preconditions	**4 sustainable policies have been passed and the final policy selection has begun.**	
Procedure	**Action**	**Expected Result**
1	The vice chooses a sustainable policy	The policy is shown as sustainable and the sustainable counter reaches 5
Expected Result	The game should end, the victorious side should be declared, the roles should be revealed, and the enacted policies should be displayed	
Actual Result	The game ends, the reformers are declared winners, the roles are revealed, and all enacted policies are displayed	
Postcondition	**The player is given the choice to go back to the main menu**	
Status	Pass	

TC024	Win Condition – Exploiters	
Preconditions	**2 exploitative policies have been passed and the final policy selection has begun.**	
Procedure	**Action**	**Expected Result**
1	The vice chooses an exploitative policy	The policy is shown as sustainable and the sustainable counter reaches 5
Expected Result	The game should end and the exploiters should be declared winners	
Actual Result	The game ends and the exploiters are declared winners	
Postcondition	**The player is on the "game over" screen**	
Status	Pass	

TC025	Game Summary	
Preconditions	**2 exploitative policies have been passed and the final policy selection has begun.**	
Procedure	**Action**	**Expected Result**
1	The vice chooses an exploitative policy	The policy is shown as sustainable and the sustainable counter reaches 5
Expected Result	The game should end and display a summary showing the winning team, all enacted policies, each player's secret role, and character selections	
Actual Result	The game ends, and a summary is displayed showing the exploiters won, all the enacted policies, who was a reformer and exploiter, and all the character selections	
Postcondition	**The player is given the choice to go back to the main menu**	
Status	Pass	

TC026	Game Persistence

Preconditions	**2 exploitative policies have been passed and the final policy selection has begun**	
Procedure	**Action**	**Expected Result**
1	The vice chooses an exploitative policy	The policy is shown as sustainable and the sustainable counter reaches 5
Expected Result	The game should end and the result is stored with a timestamp in the database	
Actual Result	The game ends and the result is stored with a timestamp in the database	
Postcondition	**The player is on the “game over” screen**	
Status	Pass	

TC027	**Round Readiness**	
Preconditions	**Policy has been enacted**	
Procedure	**Action**	**Expected Result**
1	Players clicks on “next round” button	The button changes to “waiting”
Expected Result	The game should be paused until everyone has pressed on the “next round” button	
Actual Result	The game is paused	
Postcondition	**The player waits until everyone has said they’re ready**	
Status	Pass	

TC028	**Game Reset**	
Preconditions	**Game has finished**	
Procedure	**Action**	**Expected Result**

1	Player views summary.	The player is given the option to leave to the main menu
Expected Result	After the return button is pressed, the game should be reset, game state should be cleared, and the player should be taken back to the main menu	
Actual Result	The game is reset, the game state is cleared, and the player is returned to the main menu	
Postcondition	**The player is on the menu page**	
Status	Pass	

TC029	**Disconnection Handling**	
Preconditions	**Game is ongoing**	
Procedure	**Action**	**Expected Result**
1	The player disconnects or leaves the game	The player returns to the main menu
Expected Result	The game should remove the disconnected player from the player list and notify the other players	
Actual Result	The game removes the disconnected player from the player list and notifies the other players	
Postcondition	**The player is on the menu page, the game is halted**	
Status	Pass	

TC030	**Phase Enforcement**	
Preconditions	**Vice is being nominated**	
Procedure	**Action**	**Expected Result**
Expected Result	The game should only allow the player to do what is designated to the current phase	

Actual Result	The game gives the player only one option, which is to wait until the leader nominates a vice.
Postcondition	**The game is paused until the leader selects their vice**
Status	Pass

Exploratory Testing Charters

Charter 1: Full Multiplayer Session

GOAL	**EXPLORE** the full multiplayer gameplay loop **WITH** 5–10 concurrent players **TO** analyze system stability, synchronization, and correctness of phase transitions.
TIME	Min. 30 min Max. 60 min.
SETUP	Open the deployed game: https://front-end-production-c64e.up.railway.app/ Create or join a lobby with 0-5 players
FOCUS	- Play through a full game session from lobby to game start to multiple rounds to game end - Observe phase transitions (nomination, voting, enactment) across all players. - Verify Leader/Vice rotation consistency across rounds - Check for desynchronization issues between clients - Confirm that win conditions trigger correctly and end the game

Charter 2: Edge Cases and Failure Handling

GOAL	**EXPLORE** edge case scenarios **WITH** invalid inputs, rapid actions, and disconnections **TO** analyze system robustness and error handling.

TIME	Min. 30 min Max. 60 min.
SETUP	Join a lobby with multiple players using different tabs/devices Simulate unstable conditions (refreshing, disconnecting, rapid clicking)
FOCUS	- Attempt duplicate usernames and verify rejection - Spam inputs (voting, nominations) to test rate limiting and validation - Disconnect a player during key phases (voting, policy selection) - Attempt invalid actions during incorrect phases (e.g., voting during nomination) - Observe how the system recovers from interrupted gameplay

Charter 3: Database and Data Persistence

GOAL	**EXPLORE** database interactions and persistence **WITH** completed and in-progress games **TO** analyze data integrity, fallback behavior, and reliability of stored game data.
TIME	Min. 30 min Max. 60 min.
SETUP	Open the deployed game: https://front-end-production-c64e.up.railway.app/ Play at least one full game session
FOCUS	- Complete a full game and verify that the result (winner and timestamp) is stored correctly in the database - Restart the application and check whether previously stored results remain intact - Verify that policy cards are loaded from the database at game start - Simulate database failure (if possible) and confirm fallback card deck is used - Check for duplicate entries or inconsistent data (like duplicate usernames or missing records)

Test Data Summary

Testing Area	Data Used
Players	0-5 users
Roles	Random distribution
Cards	DB + fallback
Votes	Mixed scenarios
Edge Cases	Invalid inputs, disconnects

Validation Discussion

Testing confirms that the system fully supports:

- Complete gameplay lifecycle, including:
  - Lobby creation and joining
  - Character selection and role assignment
  - Leader/Vice nomination and election voting
  - Policy selection (discard + enact phases)
  - Win condition evaluation
- Multi-phase enforcement, ensuring players can only act within valid phases
- Fairness and integrity, including:
  - Controlled role distribution
  - Single-vote enforcement
  - Randomized deck handling
- Real-time consistency, with synchronized state across all clients
- Database integration, including:
  - Policy card loading
  - Persistent storage of game results

These results demonstrate that both functional requirements (FR-01 to FR-28) and non-functional requirements (performance, consistency, security, usability) are satisfied.

Conclusion

The Greenwash serious game has been thoroughly tested using structured and exploratory approaches. The results confirm that:

- The system is functionally complete and stable
- Core mechanics accurately reflect the intended serious game objectives
- The game is ready for deployment and in-class demonstration